

Econometrics II

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Problem Set 6

Due: March 25, 2026

Question 1.

Both sets of X 's are individual-level characteristics so the coefficients on these variables need to vary by choice. There is a cost to switching labor supply in the future that is *common across all switches*. The choice in period 0 is given by LY . Once an individual retires (or chooses option 0) then that individual makes no further choices (you can verify this as if $Y_{it} = 0$ then $Y_{it+1} = 0$). The columns of Y and X_{1t} indicate observations over time, ten periods in all (so in period 10 it is a static problem). The only uncertainty from the individual's perspective is on their error term. In other words, individuals have perfect foresight about their future health. The error structure is Type 1 extreme value. **Use "dataHW6 Problem1.mat" to complete this exercise. The purpose of this problem is to estimate a simple dynamic discrete choice problem. The data contain information on the choices (the Y 's), characteristics that do not vary over time (the X_1 's), and characteristics that do vary over time (the X_{1t} 's). The context of the problem is a dynamic model of labor supply. Y takes values $\{0, 1, 2, 3\}$ reflecting retirement, flex-time, part-time, or full-time participation. The first column X_1 allows for choice specific constants, while the second column of X_1 is an indicator for gender. X_{1t} is a time-varying measure of health.**

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(a) Estimate the dynamic discrete choice model including the discount factor.

Solution.

Table 1: DDC Estimates

Parameter	
<i>Flex-Time</i>	
$\beta_{0,1}$	-0.961
$\beta_{Gender,1}$	0.368
$\beta_{Health,1}$	0.517
<i>Part-Time</i>	
$\beta_{0,2}$	-.051
$\beta_{Gender,2}$	0.687
$\beta_{Health,2}$	0.759
<i>Full-Time</i>	
$\beta_{0,3}$	-0.412
$\beta_{Gender,3}$	1.029
$\beta_{Health,3}$	0.992
δ	0.376
β	0.896

(b) What variation in the data allows the discount factor to be identified?

The discount factor is identified via the health state. It is the only time variant characteristic in X_{1t} , so it will pin down β .

(c) What variation in the data allows the switching cost to be identified?

Switching cost is identified via the constant, the only characteristic which is choice specific in X_1 . Gender is not something which we are assuming can change in this model, thus it must be the constant identifying the cost δ .